Computer Communication and Networks

Physical Layer

Session 5

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The Physical Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Describe different transmission media.
- Identify the guided media types.
- Explain each type of guided media
- Define unguided media
- Identify wireless communication category



The Physical Layer: Introduction

Introduction:

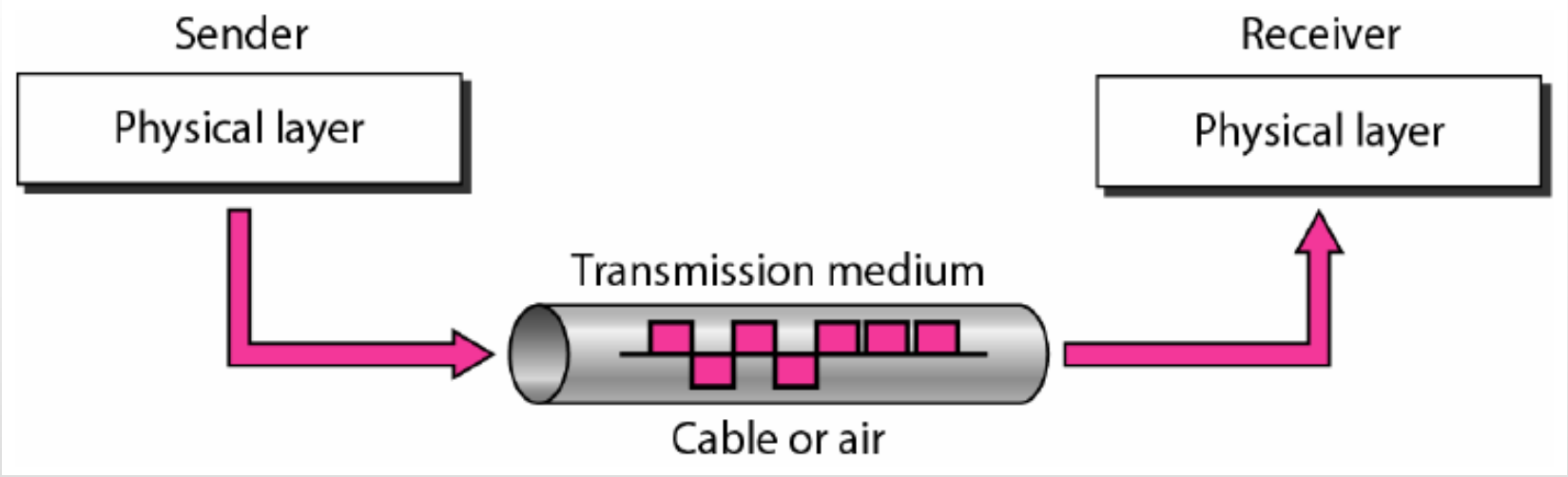
- Physical layer is the lowest layer in our protocol model.
  - It defines the **electrical**, **timing** and other interfaces by which bits are sent as signals over channels.
  - It is the foundation on which the network is built.
  - It provides the basic communication channel that two network devices use to sent and receive messages



The Physical Layer: Introduction

Introduction:

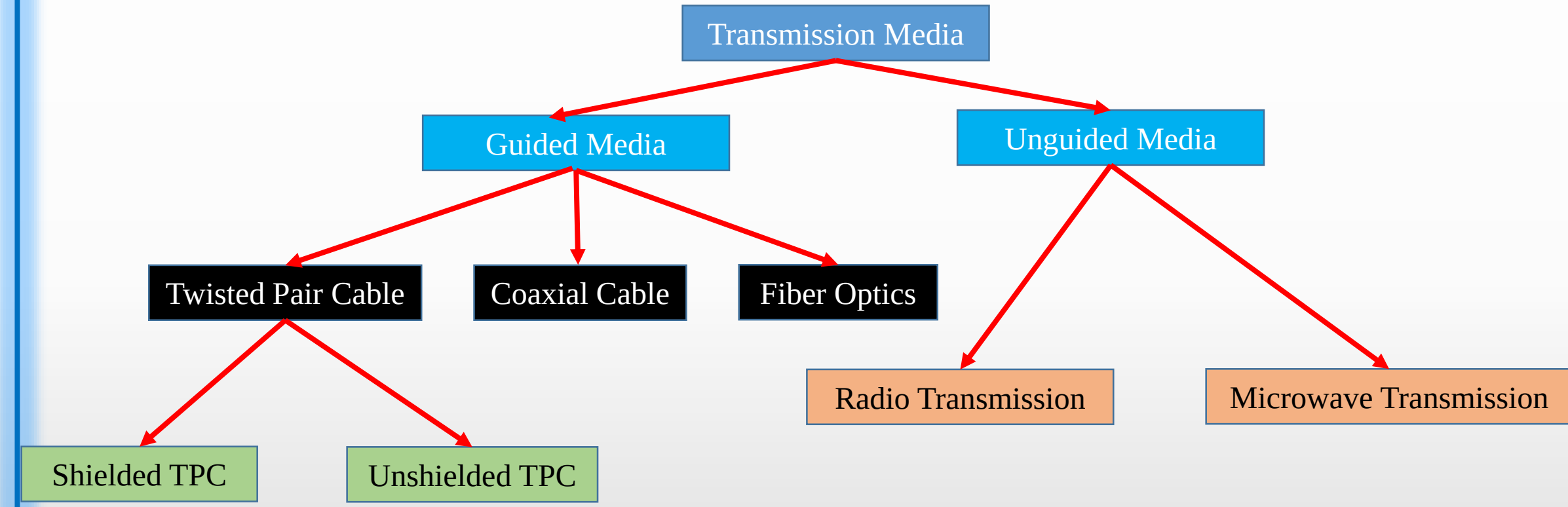
- The transmission media is located below the physical layer and directly control by the physical layer.
- The figure below shows the location of transmission media in relation to physical layer.





The Transmission Media: classification

❖ Types Transmission Media:





The Physical Layer: Transmission media factors

❖ Transmission Media:

➢ Factors to be considered while selecting Transmission Media:

1. Transmission Rate
2. Cost and East of Installation
3. Resistance to Environmental Condition
4. Distance



The Physical Layer: The transmission Media

❖ Transmission Media:

- There are two types of transmission media:
  1. Guided (copper wire and fiber optics).
  2. Unguided/Wireless (terrestrial radio).

  - Each of these technologies has different properties that affect the design and performance of the networks that use them.



The Transmission Media: classification

Transmission Media:

Guided

- It is the transmission media in which signals are confined and directed to a specific path using wire or cable.

- ✓ Three basic guided transmission media are:
  1. Twisted pair cable
  2. Coaxial cable.
  3. Optical fiber.

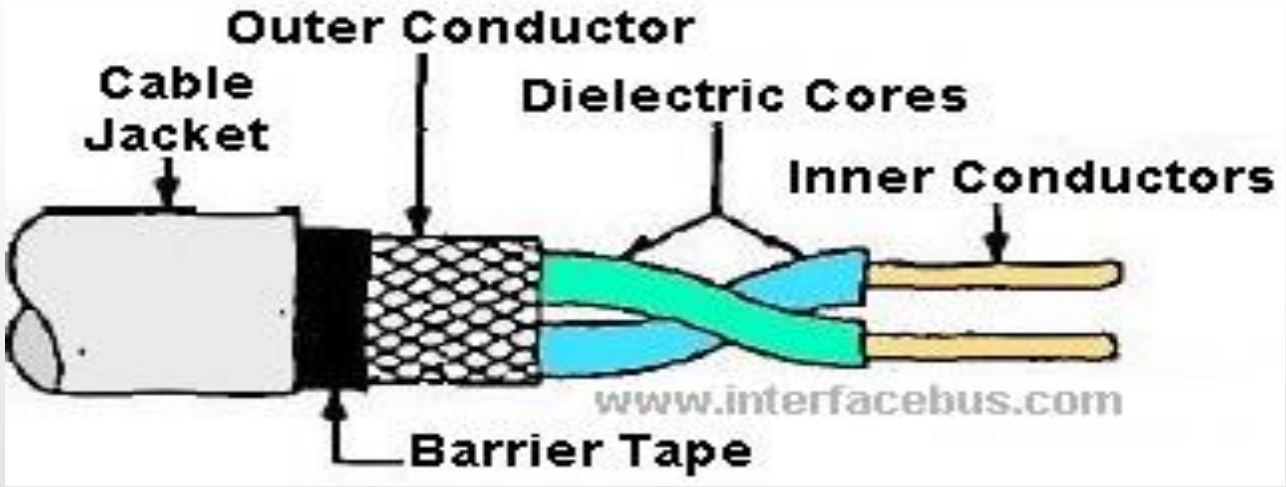


The Transmission Media: Guided

❖ Guided: Twisted Pair Cable

➢ What is Twisted Pair Cable?

- ✓ A twisted pair cable consists of two conductors (copper), each with its own plastic insulation, twisted together as shown in the picture below:



- ✓ One of the wires is use to carry a signal to the receiver and other is only for ground reference.



The Transmission Media: Guided

❖ Guided: Twisted Pair Cable

➢ Why this cable is Twisted?

- ✓ The purpose of twisting in twisted pair cable is to remove or canceling out the Electromagnetic Interference (EMI) from the external sources.

➢ Two types of Twisted Pair Cable:

1. Unshielded Twisted Pair Cable (UTP)
2. Shielded Twisted Pair Cable (STP)



The Transmission Media: Guided

❖ Twisted Pair Cable: Advantages and disadvantages

➢ Twisted pair cable are the oldest and most popular cables all over the world. This is due to the many advantages that they offer:

1. Trained personnel easily available due to shallow Learning curve
2. Can be used for both analog and digital transmissions
3. Least expensive for short distances
4. Entire network does not go down if a part of network is damaged.



The Transmission Media: Guided

❖ Twisted Pair Cable: Advantages and disadvantages

➢ Disadvantages:

1. Signal cannot travel long distances without repeaters
2. High error rate for distances greater than 100m
3. Very thin and hence breaks easily
4. Not suitable for broadband connections

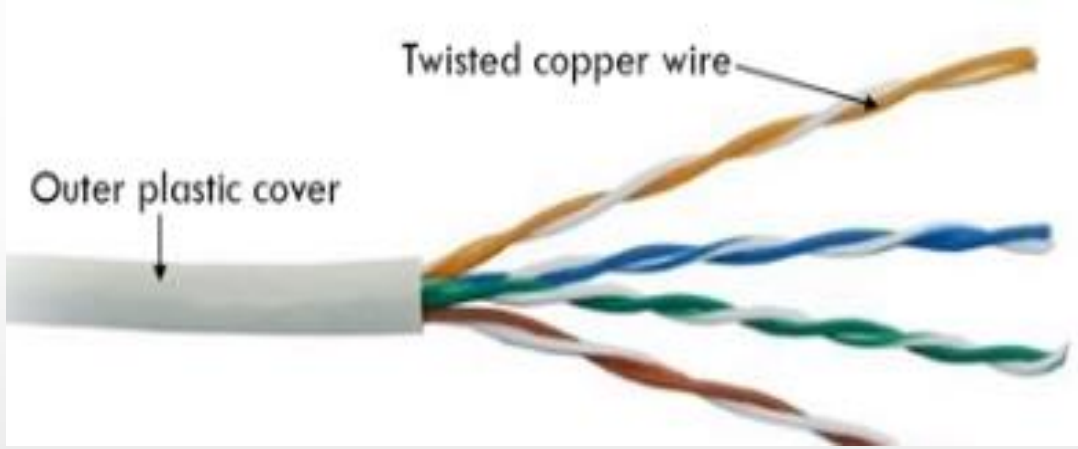


The Transmission Media: Twisted Pair

❖ UTP and STP: Two types of Twisted Pair cables.

➢ Unshielded Twisted Pair:

- ✓ In unshielded twisted pair cable, the shield is not provided. Copper wires are twisted with colour plastic insulator



- ✓ Most commonly used twisted-pair
- ✓ Electronic Industry Association (EIA) developed to classify UTP viz. CAT 1 – CAT 8.
- ✓ Categories are determined by cable quality(1-lowest, 8-highest)



The Transmission Media: Twisted Pair

❖ UTP and STP: Two types of Twisted Pair cables.

➢ Shielded Twisted Pair:

- ✓ STP has a metal foil or braided mesh covering each pair of insulated conductor.
- ✓ Metal casing improves the quality of cable by preventing the penetration of noise or crosstalk but it is bulkier and more expensive.





The Transmission Media: Twisted Pair

❖ Comparison between UTP & STP

Sr. No.	Parameter	UTP (Unshielded)	STP (Shielded)
(1)	Data Rate	10-100 Mbps	150 Mbps
(2)	Cable length	100 meter max	500 meter max
(3)	Electrical interference	most Interference and cross talk occurs	Less Interference and cross talk occurs
(4)	Installation	Easy to install	Very easy to install
(5)	Cost	Lowest	Little costly

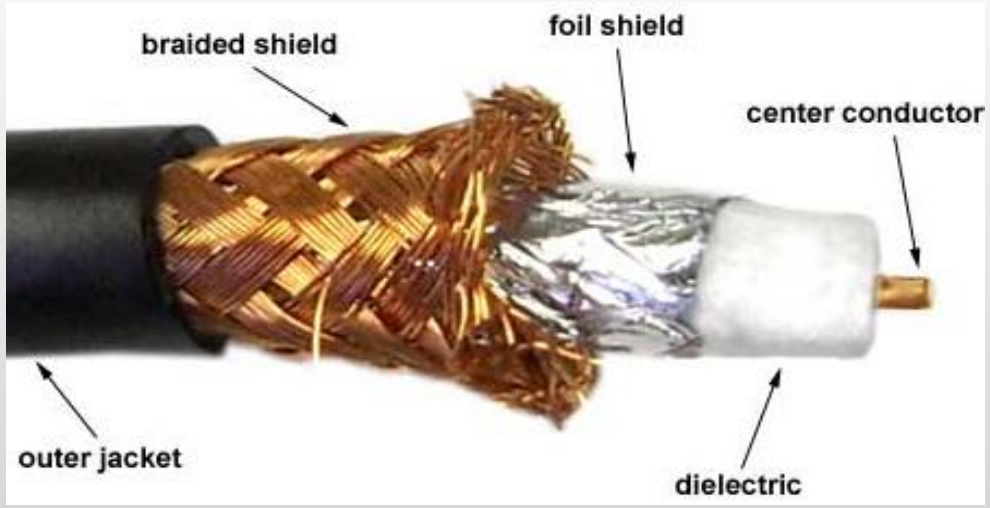


The Transmission Media: Guided

❖ Coaxial Cable:

➢ Coaxial Cable:

- ✓ Coaxial Cable is a cable which has two conductors that are parallel to each other. Hence, this cable is called as coaxial, means at the same axis.
- ✓ It consists of **copper core** surrounded by **insulated material** and a **braided outer conductor**.
- ✓ Carries signal of higher frequency ranges than those in twisted-pair
- ✓ Coaxial cable used in telephone system is replaced by fiber.





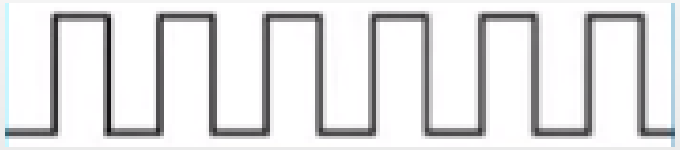
The Transmission Media: Guided

❖ Coaxial Cable:

➢ Two types Coaxial Cable:

1. Baseband
2. Broadband

➢ Baseband:



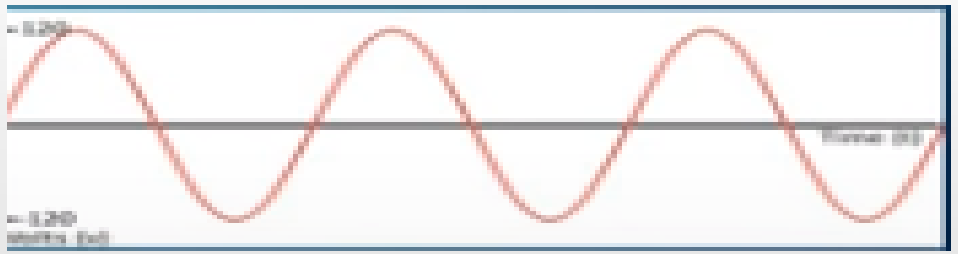
1. It is use for transmission of digital signal
2. It is mostly used for LANs
3. It transmits a signal at a time with very high speed
4. The major drawback is that it needs amplification after every 1000 feet.



The Transmission Media: Guided

❖ Coaxial Cable:

- ➢ Broadband:
  1. It is use for transmission of analog signal
  2. It is mostly used for TV
  3. It transmits several simultaneous signal using different frequencies
  4. It covers large area



❖ Guided Transmission Media:

➢ Coaxial Cable Standard

- ✓ Coaxial cables are categorize by Radio Government (RG) rating.

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Baseband (digital)

10 Mbps

Maximum length

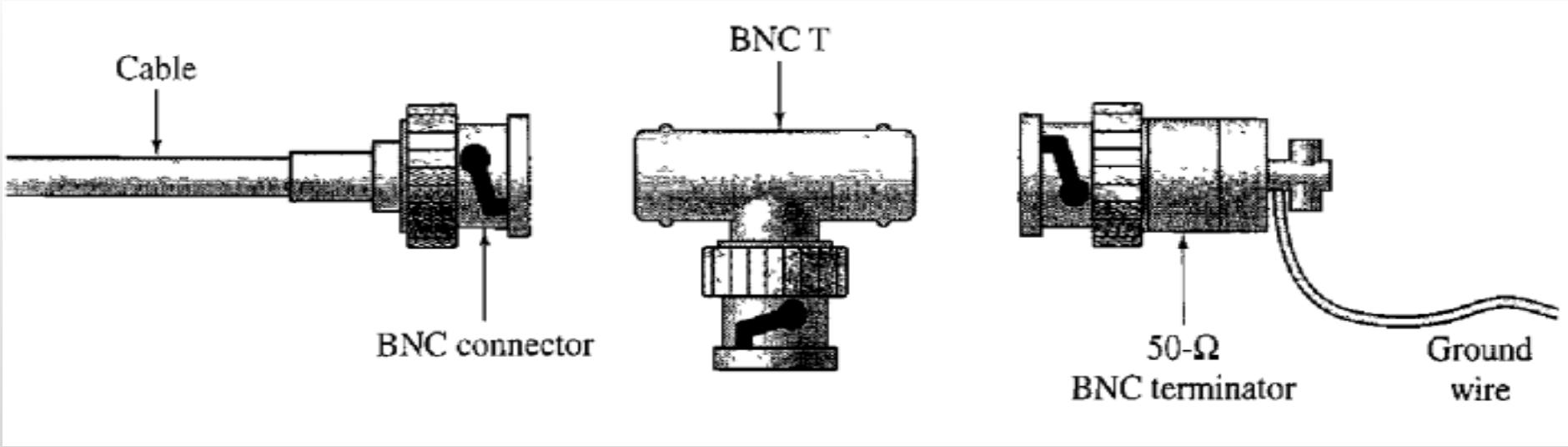
- ✓ 10Base2
- ✓ Uses thinner coaxial cable

- ✓ 10Base5
- ✓ Uses thicker coaxial cable

❖ Guided Transmission Media:

➢ Cable connectors:

- ✓ To connect coaxial cable to devices, we need connector.
- ✓ The most common type of connector use today is BNC(Bayone-Neill-Concelman) connector.
  - 3 types: BNC Connector, BNC T & BNC Terminator





The Transmission Media: Guided

❖ Guided Transmission Media:

➢ **Fiber Optics Cable: *Optical Fiber***

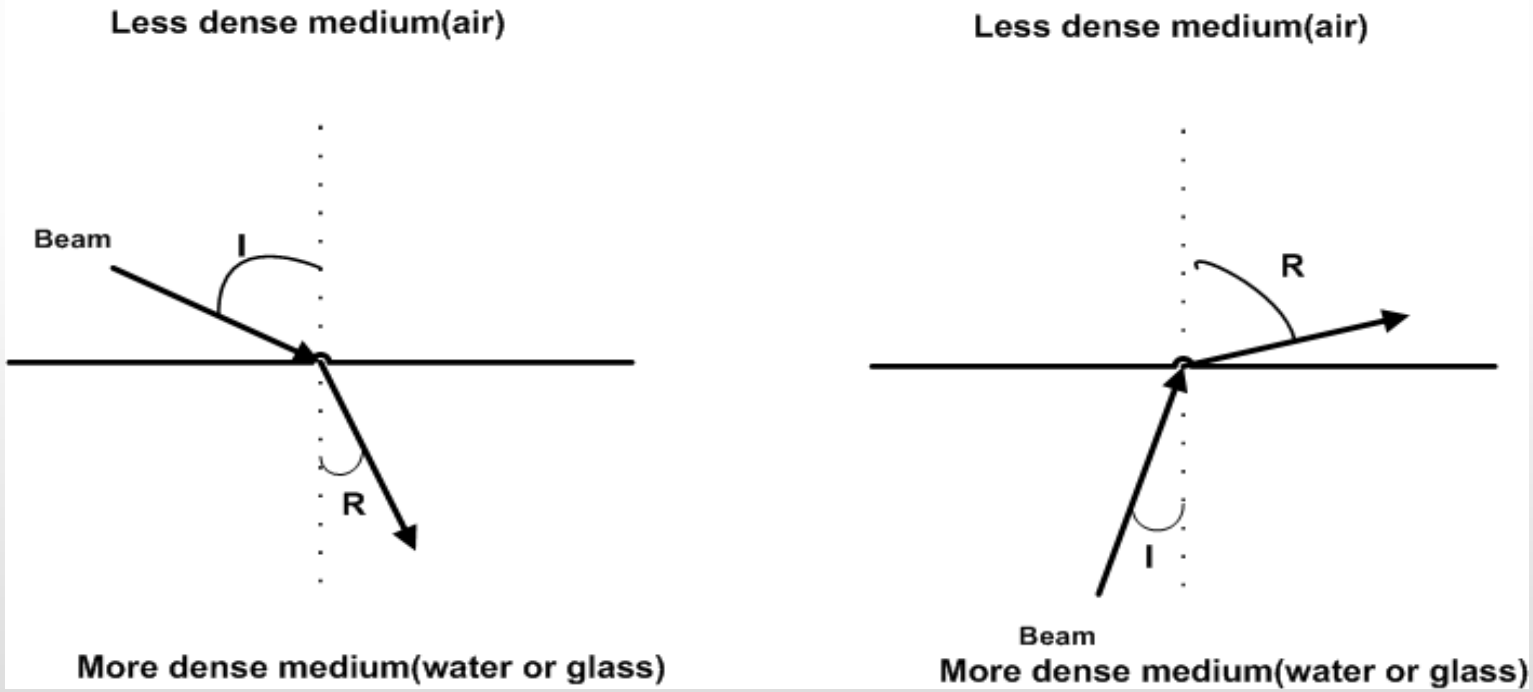
- ✓ Optical fibers use **reflection** to guide light through a channel.
- ✓ Fiber-Optic cable is either made up of **glass** or **plastic**
- ✓ It transmits signal in the form of light.
  - The light is a form of **electromagnetic energy**.
- ✓ The speed of light depends on **the density of the medium through which it is traveling.**
- ✓ The higher the density the slower the speed



The Transmission Media: Guided

Guided Transmission Media:

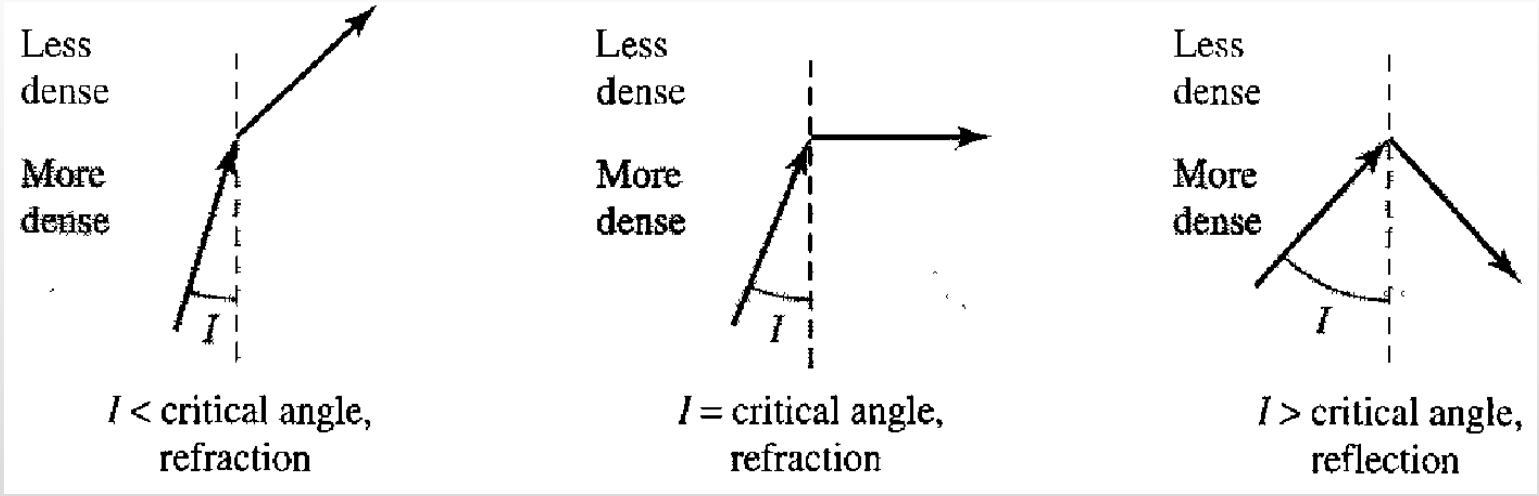
- When light travels into a more dense medium, the angle of incident(I) is greater than the angle of refraction(R); and when it travels into a less dense medium, the angle of incidence is less than the angle of refraction.



❖ Guided Transmission Media:

➢ Critical angle:

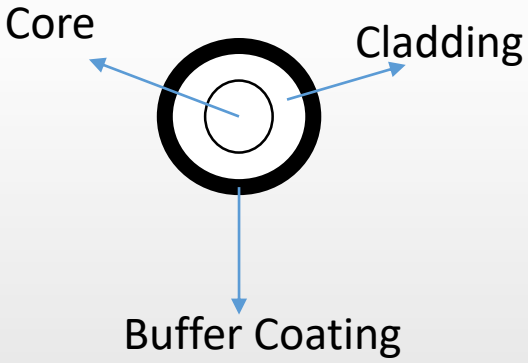
- ✓ When incident angle results in a refracted angle of 90 degrees (i.e refracted beam lying along the horizontal). The incident angle at this point is known as the critical angle.
- ✓ Reflection occurs when the incident angle is more than the critical angle; otherwise refraction occurs as shown in the following.

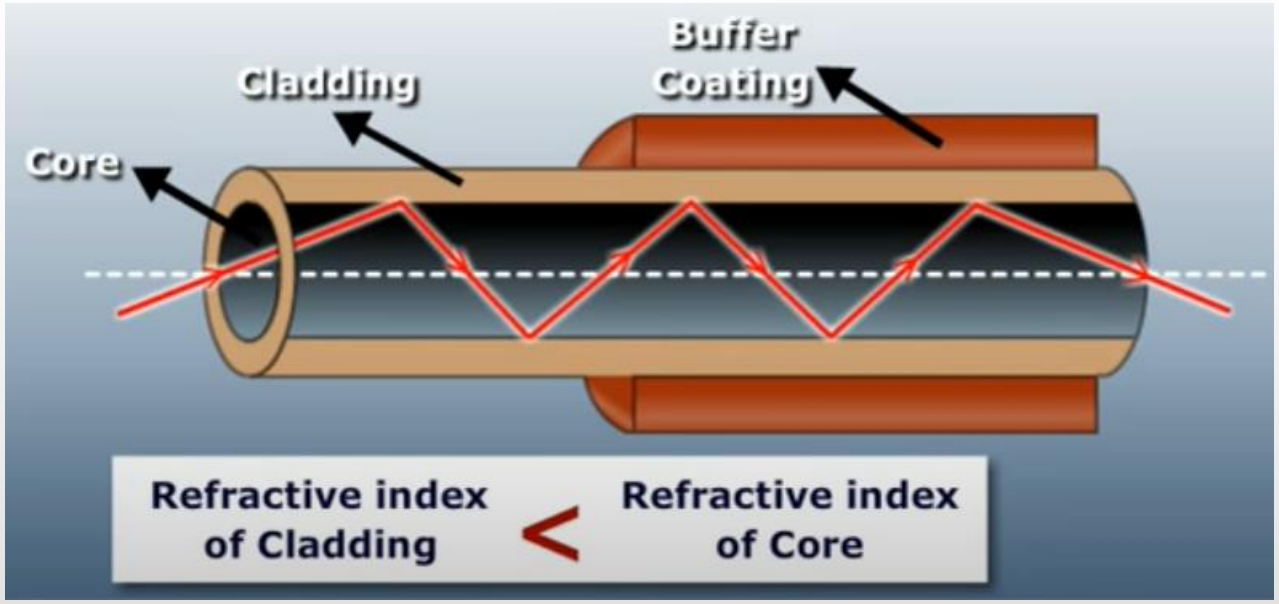


❖ Optical Fiber

➢ Structure of the optical fiber

✓ It consists of three concentric sections:







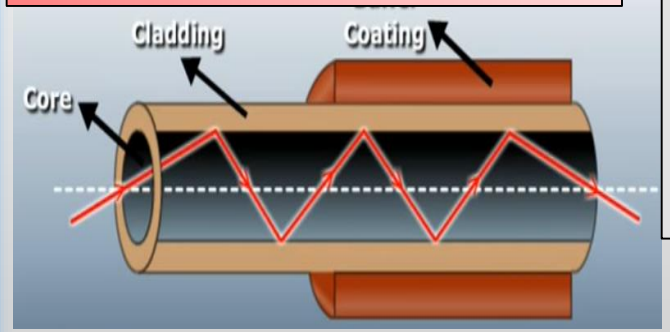
The Transmission Media: Guided

Optical Fiber

- **Explanation of the structure of the optical fiber**
  - Optical fiber is the thin flexible medium and is cylindrical in shape.
  - It consists of Core, **Cladding**, and Buffer coating.

The Core is the inner cylinder which is made of either glass or plastic.

It is the light guiding region of the fiber.



The **Cladding** is anther cylindrical shape which surrounds the core.

It is also either made of glass or plastic.

The refractive index of cladding is lower than the refractive index of the core.

Therefore, the cladding keeps the light guided within the core.

The Buffer Coating is the outer most region of the optic fiber.

It is a plastic or polymer coating given to the cladding to protect against moisture or crushing operation and the other environmental hazard.

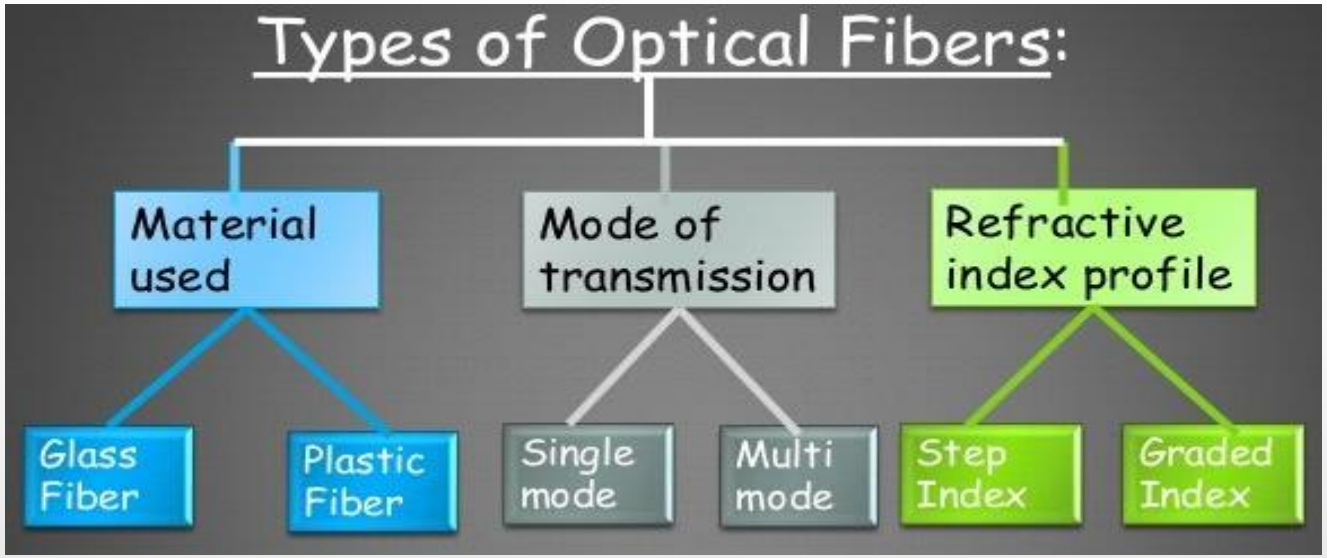


The Transmission Media: Guided

❖ Types of Optical Fiber

➢ There are three types of optical fiber:

1. Single Mode
2. Multimode Step Index
3. Multimode Graded Index





The Transmission Media: Guided

❖ Guided Transmission Media:

- **Fiber sizes:**
  - ✓ Optical fiber is defined by the **ratio** of the diameter of their core to the diameter of their cladding, both express in micrometers.

Type	*Core (μm)*	*Cladding (μm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Feature	Single Mode Optical Fiber	Multimode Step Index Optical Fiber	Multimode Graded Index Optical Fiber
Core Diameter	8-10 micrometers	50-200 micrometers	50-100 micrometers
Mode of Propagation	Single mode (one light path)	Multiple modes (multiple light paths)	Multiple modes with varying path lengths
Refractive Index Profile	Step index (uniform core, abrupt change at cladding)	Step index (uniform core, abrupt change at cladding)	Graded index (refractive index gradually decreases from core center to edge)
Bandwidth	Very high	Lower than single mode	Higher than step index multimode
Distance	Long distances (up to 100 km)	Short distances (up to 2 km)	Medium distances (up to 2 km)
Dispersion	Low	High	Moderate
Light Source	Laser (typically)	LED or Laser	LED or Laser
Applications	Long-haul telecommunication, high-speed networks	Short-distance data communication, LANs	Medium-distance data communication, LANs
Cost	Higher	Lower	Moderate
Installation Complexity	Higher	Lower	Moderate



The Transmission Media: Guided

❖ Frequently Asked Questions

1. What is Dispersion?

Dispersion in optical fibers refers to the spreading of light pulses as they travel down the fiber.

2. What is Attenuation?

Signals lose strength as they are propagated through the fiber; this is known as an attenuation.

Attenuation is measured in decibels (dB)



The Transmission Media: Guided

❖ Frequently Asked Questions

3. What is used for the fabrication of optical fibres that are used for communication?

   Silica or multi-component glass are used for the fabrication of optical fibres.

4. Why is silica used for the fabrication of the optical fibres?

   Silica has a perfect elasticity until it reaches the breaking point, which makes it best for fabrication.



The Transmission Media: Guided

❖ Frequently Asked Questions

5. What is the principle of fibre optical communication?

Total internal reflection is the principle on which the optical fibre communication is based.

6. What is the Raman effect?

The Raman effect is the change in the wavelength of light which occurs when the molecules deflect the light beam.



❖ Frequently Asked Questions

7. Name the factors that are responsible for generating attenuation of optical power in fibre.

Following are the factors that are responsible for generating attenuation of optical power in fibre:

1. Absorption
2. Scattering
3. Waveguide effect



The Transmission Media: Guided

❖ Frequently Asked Questions

8. Why are plastic-clad silica fibre optic cables not user-friendly?

Following are the reasons why plastic-clad silica fibre optic cables not user-friendly:

1. The fibres are insoluble in organic solvents
2. Bonding becomes difficult
3. Connector application becomes difficult as there is excessive plasticity in the cladding



The Transmission Media: Guided

❖ Frequently Asked Questions

9. What are the benefits of optical fibre cable?

Following are the benefits of optical fibre cable:

1. The data security is excellent
2. It is cost-effective
3. It won’t be affected by interference



The Transmission Media: Guided

❖ Frequently Asked Questions

10. What is the bandwidth of optical fibre?

    900 THz is the bandwidth of the optical fibre.

11. What is core in the optical fibre?

    The core is the inner part of the optical fibre, which carries light.



The Transmission Media: Guided

❖ Frequently Asked Questions

12. What are the benefits of optical fibre cable?

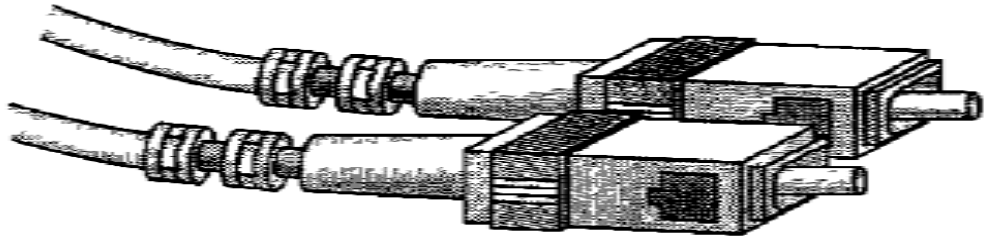
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❖ Fiber-optic connector:

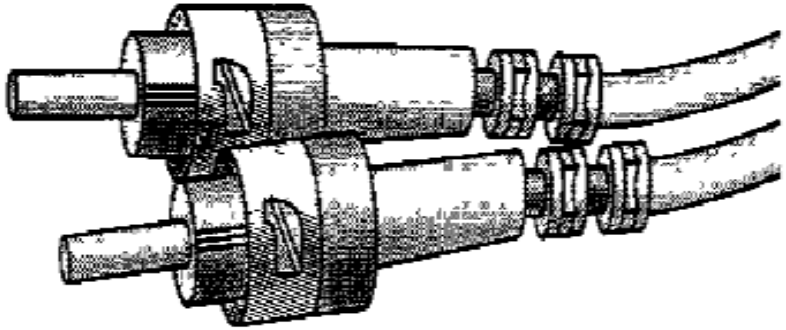
➢ **3 types fiber-optic Connectors:**

1. **Subscriber Channel (SC) Connector**
   - ✓ It is used for TV cable



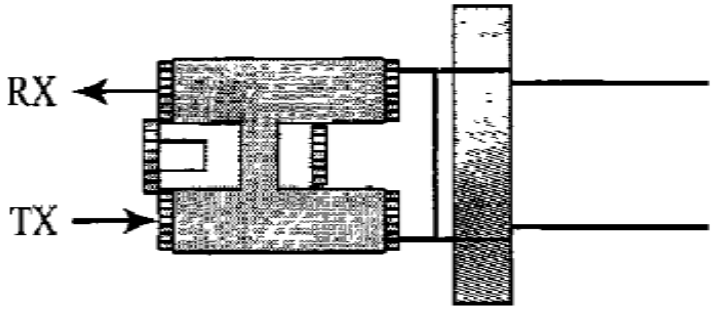
SC connector

2. **Straight-tip (ST) Connector**
   - ✓ It is used to connect cable to networking devices



ST connector

3. **MT-RJ Connector**
   - ✓ Same size as RJ45



MT-RJ connector



The Transmission Media: Guided

❖ Advantages of Optical fiber:

➢ Fiber-optic cable has several advantages over metallic cable.

1. **Higher bandwidth**
2. **Less signal attenuation**:- The signal can run for 50Km without requiring regeneration.
3. **Immunity to electromagnetic interference(Noise** resistance): Fiber optic uses light rather than electricity, noise is not a factor.
4. **Light weight**
5. **Greater immunity to tapping:** more immune to tapping than copper cable. Copper creates antenna effects that can easily be tapped.



The Transmission Media: Guided

❖ Disadvantages of Optical fiber:

- Fiber-optic cable has disadvantages.

1. Installation and maintenance
2. Unidirectional light propagation
3. Cost



References

References:

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4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan